



Game System & Gameplay Development & Framework Maintenance
2025.08 - 2025.11

Yunmengshan Academy (AgentLand Lab, Tsinghua SIGS): Architected the core systems for **Yunmengshan Academy**, an AI-driven management sim featuring LLM-integrated NPCs. Engineered the end-to-end card progression loop—encompassing collection, inventory, and combat—and spearheaded the "Lundao" debate mode by embedding **LLM** logic into the battle engine. Maintained the project's foundational framework to ensure architectural stability and seamless system integration.

LLM

Card Combat

Agent Progression

01 YUNMENGSHAN ACADEMY

Gameplay Flow - 1

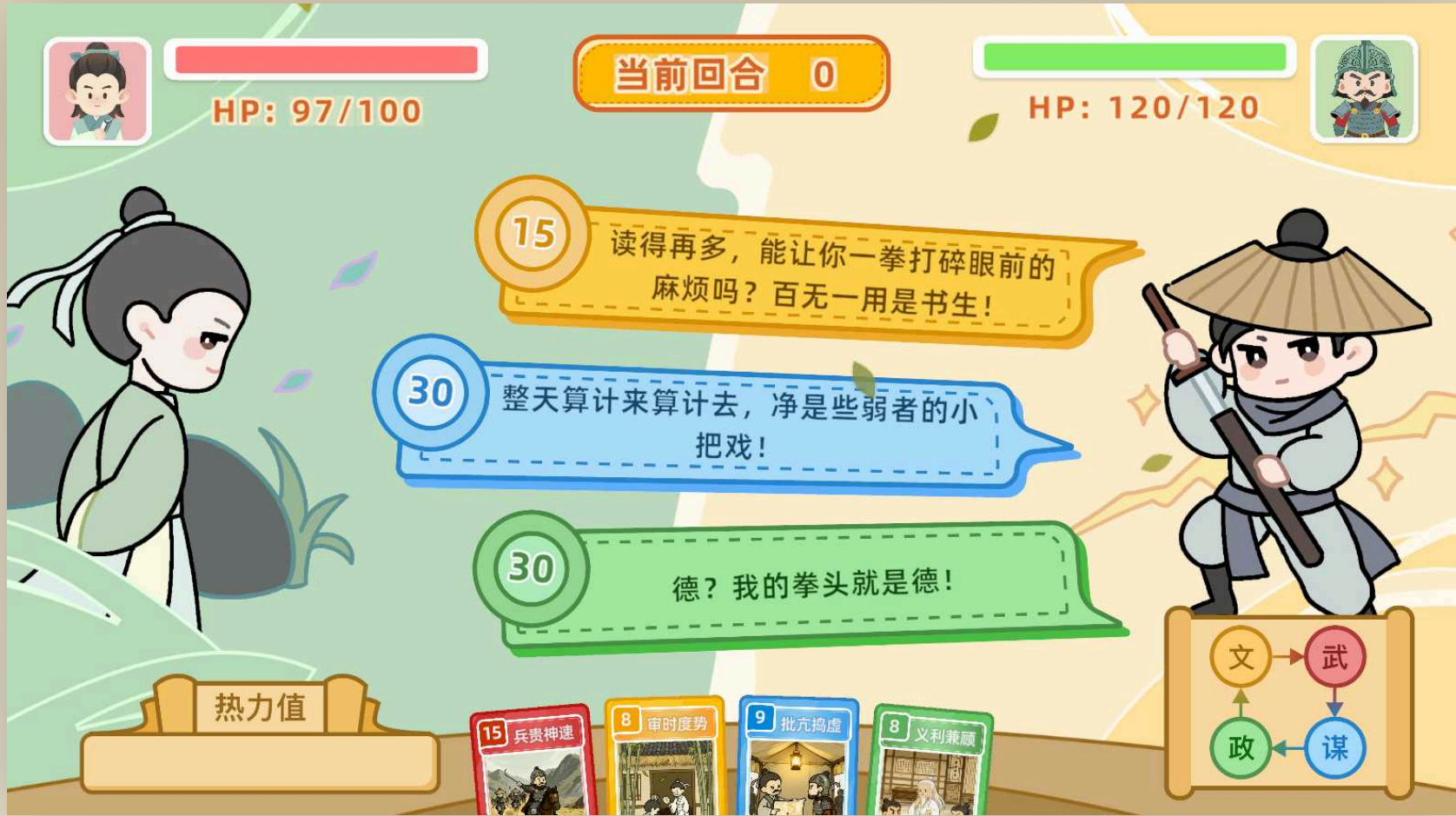
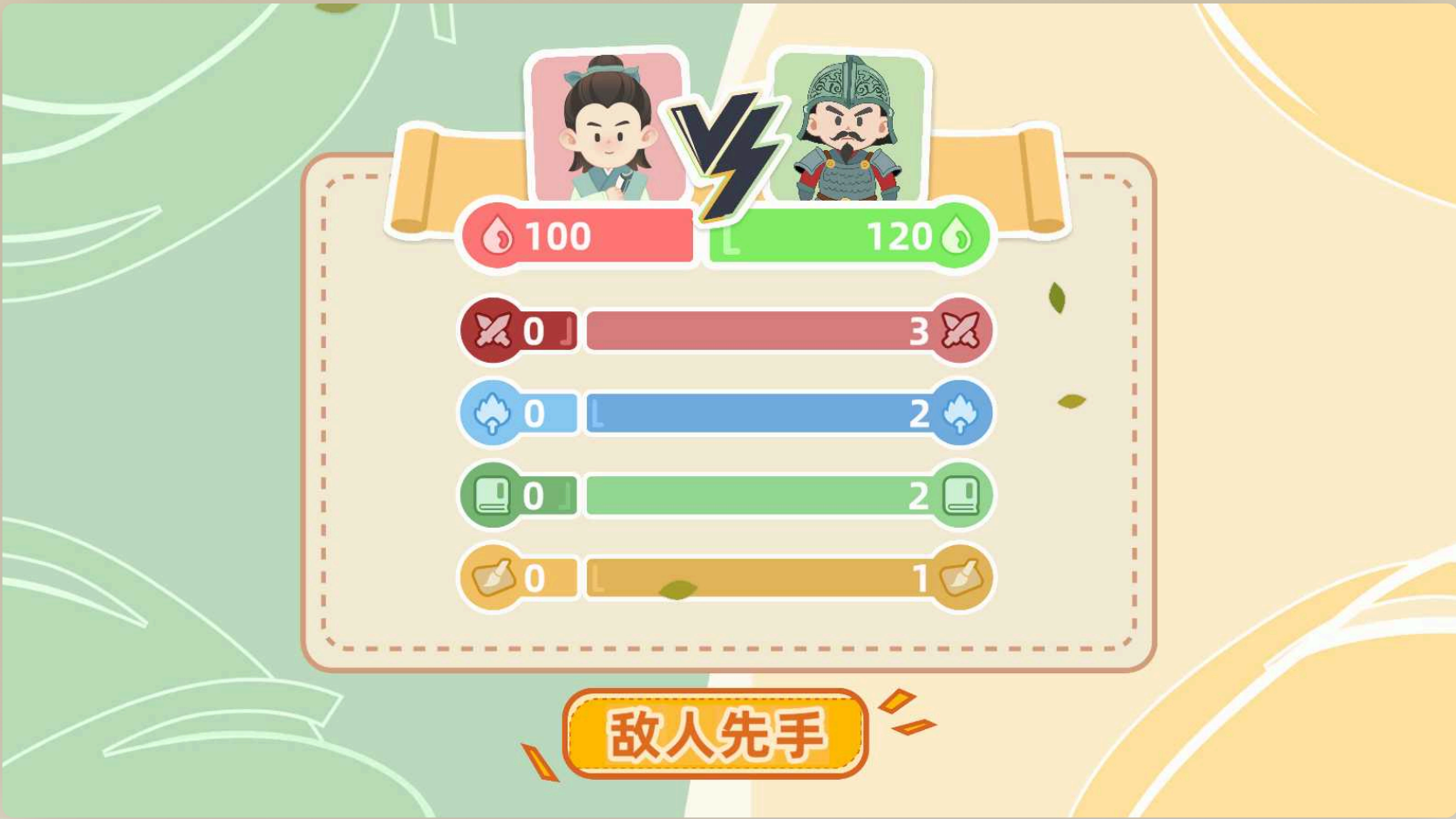


Card Collection

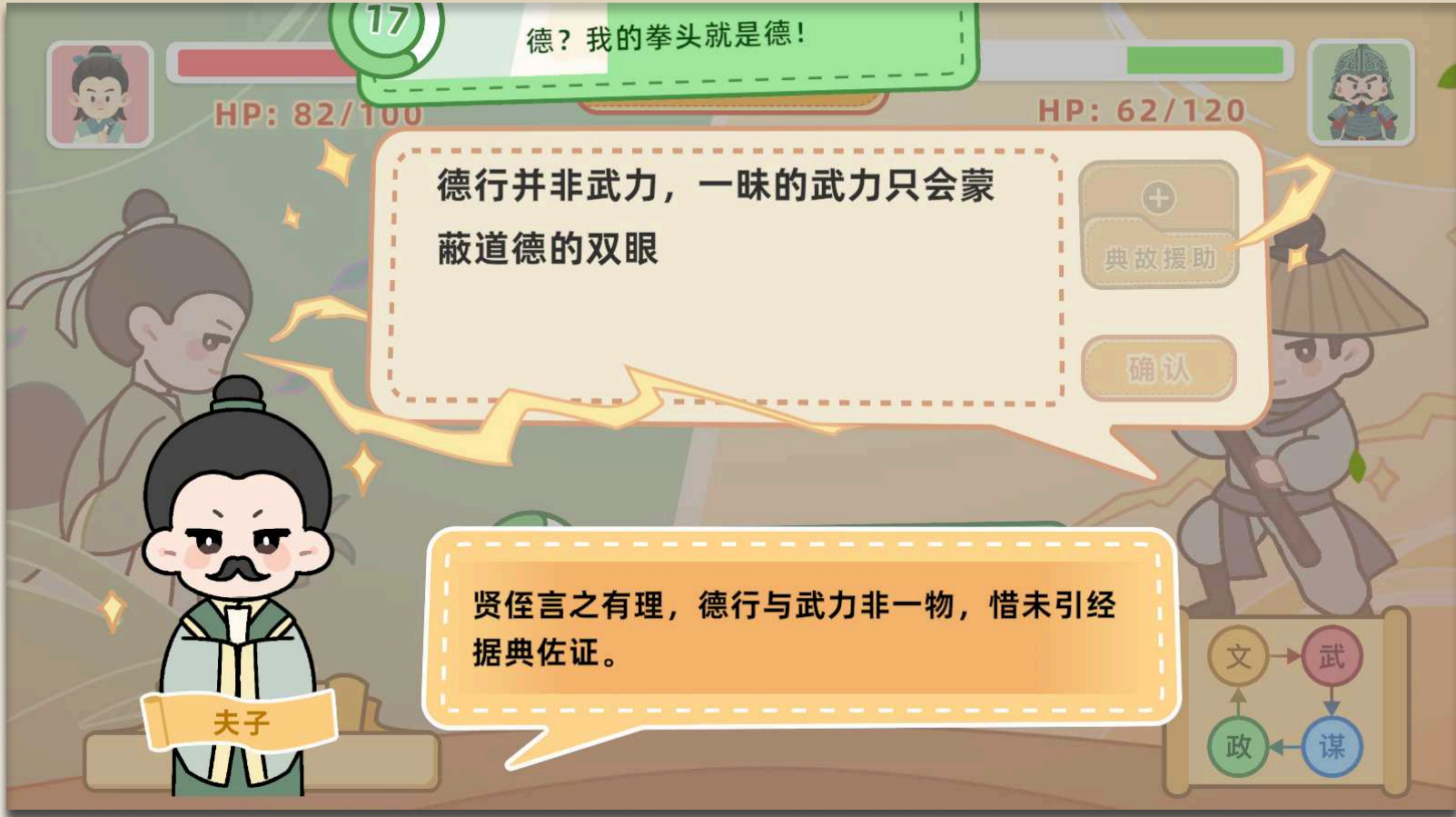
1 Cards collected during gameplay are stored in an inventory where players can **manage and customize** them. I designed and implemented the full functionality of the **card system** and **inventory interface**, ensuring alignment with the game's setting.

Card Combat

2 Players enhance four core attributes by completing main and daily quests. These attributes are compared against the enemy's before card battles; the side with superior stats gains the initiative.



Gameplay Flow - 2



Fever Attack

3 Developed the **Fever Attack** mechanic, where an LLM generates high-damage output based on the debate background and viewpoints. The system integrates player inputs with classical allusions and utilizes a predefined prompt to dynamically calculate the damage multiplier.



DebateMode

4 Building upon the aforementioned card combat, there is an alternative mode known as 'Debate' (Lun Dao). In this mode, players are provided with a specific set of cards and must identify fallacies in the opponent's arguments according to the mini-game mechanics. By using their cards to refute these viewpoints, players influence the NPC, who will then adjust their strategy based on the outcome.

Technology & Solutions

Card

Deck

Enemy

Debate

Curve

Viewpoint

EndGame

Player

ScriptableObject

All data files for the card system are stored as **ScriptableObjects** to streamline configuration and debugging for game designers. I also integrated the card system into the game's Model layer, enabling real-time data modification and debugging during runtime by updating the Model's values

SO Creator

选择 SO 类型后创建实例。所有嵌套的 SO 字段将自动展开，填写完成后一键保存所有资源。

1. 选择 ScriptableObject 类型

仅显示自定义 SO 命名空间前缀 CardGame

SO 类型 DebateData (CardGame)

找到 4 个可用类型

2. 配置保存路径和名称

保存路径 Assets/Game/Runtime/ScriptableObjects

资源名称 Debate2

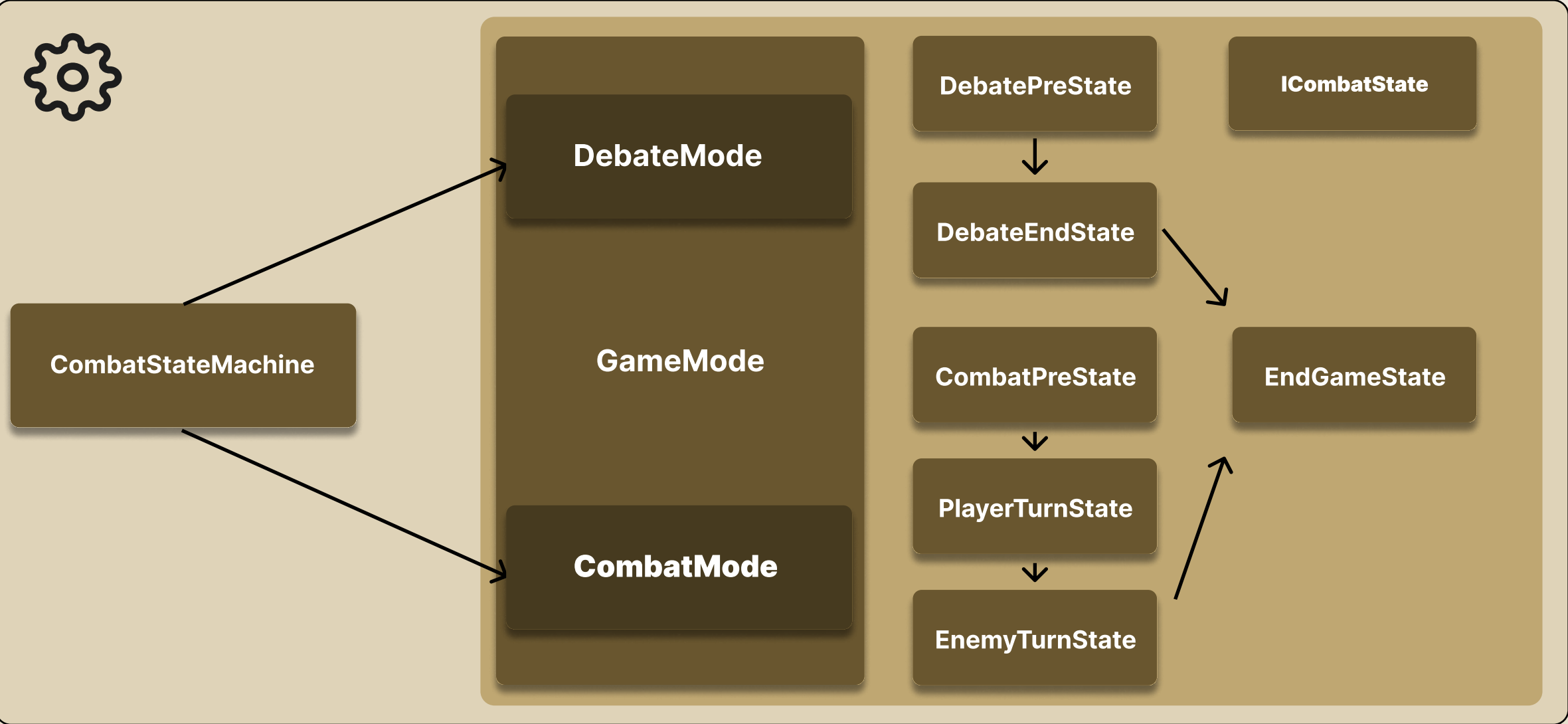
3. 创建实例并填写数据

创建 ScriptableObject 实例

4. 编辑实例数据

Custom Unity Editor tool for rapid generation of hierarchical and complex ScriptableObjects

Finite State Machine



- **Logic Decoupling:** FSM ensures seamless transitions between Combat and Debate modes by decoupling complex states.
- **State Management:** Encapsulated sub-states allow precise control over animations and UI.
- **Interface-Driven:** Utilizes ICombatState for high cohesion and low coupling, enhancing maintainability.

Prompt Engineering

典故摘要提示词

Summary Prompt

#Role: 辩论文段处理机器

##Profile

-Language: 中文

-Description: 你需要扮演一个春秋战国时期的协助辩论产出文段的机器，针对某一待反驳的观点，根据玩家输入的反驳语句，和玩家想融合引用的典故，重新结合输出一段有理有据的反驳文段。

##ViewPoint

玩家正在与一位对手进行辩论，这位对手的人设是：(NPC的人设和观点)

本次需要反驳的对手观点是：(待反驳观点)

##Player Retort

玩家自行输入的反驳语句是：(玩家本次的输入)

##Classical reference

玩家希望引用到的典故是：(选择的典故)

热力提示词

Fever Prompt

#Role: 思想辩论评分机器

##Profile

-Language: 中文

-Description: 你需要扮演一个春秋战国时期的思想辩论裁判，负责评估“玩家反驳语句”对“反驳NPC观点”的辩论伤害等级，并判断“玩家选择的典故”用于反驳“反驳NPC观点”是否合适。你需要根据一定的评级标准返回合适的评级数字，并根据玩家引用典故的情况返回典故是否正确。

##Rule

以下是你需要参考的评级标准，一共有3条:

1.关系克制: 你需要将玩家的言论分四类属性，将其和NPC属性进行克制对比。

> - **文**: 涉及礼乐、教化、典籍 (如引用《诗》《书》)

> - **武**: 涉及暴力、刑罚、军事压制

> - **谋**: 涉及权术、法令、操控人心

评论提示词

Comment Prompt

Role: 卡牌辩论对话生成器

Profile

- Language: 中文

- Description: 你需要扮演一个春秋战国时期的思想辩论裁判，负责生成“玩家打出卡牌”后对“反驳NPC观点”的辩论对话。你需要为卡牌游戏返回两句简短对话：玩家的反驳和敌人的回应。

敌人攻击提示词

Enemy Attack Prompt

Role: 敌人观点攻击对话生成器

Profile

- Language: 中文

Key Prompts

- 1.**Allusion Synthesis Prompt**
Combines player input and classical references into a structured rebuttal.
- 2.**Fever Evaluation Prompt**
Assesses the damage of player arguments on the NPC's stance
- 3.**Debate Reaction Prompt**
Generates follow-up dialogue after a card is played
- 4.**Enemy Strike Prompt**
Provides 4 pairs of enemy attacks and player counter-arguments

Shader & DOTween & Partical System

长平之战

多路并举

兵无常势

项羽溃败

围魏救赵

破釜沉舟

背水一战

Curved Card Layout

热力值

Shader-based Fire VFX

Partical-based VFX